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Potentialiation of Radiation-Induced Bacterial Cell Killing by Binuclear Rhodium(II) Complexes and Their Amines

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Rhodium(II) binuclear complexes were surveyed for potentialiation of radiation-induced cell killing of hypoxic and fully oxic *Salmonella typhimurium* cells. The Rh₂ tetracarboxylate ion substantially potentialiated hypoxic cell radiation sensitivity. Phosphate interfered with this potentialiation. In the latter two respects, radiation potentialiation by Rh₂ tetracarboxylate is similar to that found for Rh₂ tetracarboxylates. Amines such as ammonia, methylamine, ethylamine, *n*-propylamine, and *n*-butylamine were examined with both Rh₂ tetracarboxylate and tetraacetate complexes. With Rh₂ tetraacetate in phosphate-buffered saline, these amines variably increased radiation potentialiation to a maximum of nearly that seen by Rh₂ tetraacetate alone in the absence of phosphate. With Rh₂ tetracarboxylate, particular amines were found to either enhance or restrict radiation potentialiation. Results as a whole support the hypothesis that a radiolytic Rh species initiated in a one-electron reduction process external to the cell is responsible for the potentialiation by Rh₂ complexes in bacteria. Phosphate interference of potentialiation by Rh₂ tetraacetate appears to be limited competitively by amines, suggesting that axial associations of phosphate with the Rh₂ center may be involved in the inhibition of radiation potentialiation. Of interest in this regard is the finding that 5'-adenosinemonophosphate eliminates the potentialiation seen with Rh₂ tetraacetate. © 1989 Academic Press, Inc.

INTRODUCTION

Rhodium is a versatile metal for structure-activity studies pertaining to the potentialiation of radiation-induced cell killing. Surveys on the potentialiation of radiation-induced cell killing by Rh(II)₂ carboxylates have been done with mammalian (1) and bacterial (2) cells. From these surveys, thiol depletion in mammalian cells and radiolytic product formation in bacterial cells have been proposed to be involved in this radiation potentialiation.

The oxidative involvement of Rh(II)₂ carboxylates and of *cis*-dichlorodiammineplatinum(II), *cis*-DDP (3), in free radical processes during irradiation would seem limited to reaction with highly reductive radical anions, such as the hydrated electron, e_{aq}^- (1-4), since approximately -1 V one-electron reduction potentials are reported for Rh₂ carboxylates and *cis*-DDP (1). In this regard, potentialiation of radiation-induced killing of the bacterial spore by *cis*-DDP suggested that reaction with

e_{aq}^- could be important in the mechanism of potentiation (5), suggesting a Pt(I)-based process as one possible explanation for the hypoxic potentiation that has been observed (4–6). Therefore, a radiolytic active product may also contribute to the radiation potentiation seen for *cis*-DDP.

The work here extends the study of radiation potentiation by Rh₂ carboxylates (2) to a series of Rh₂ tetraacetate amines and the Rh₂ tetracarboxylate anion.

METHODS

Chemicals. Rh₂(C₂H₃O₂)₄, rhodium tetraacetate, was provided by Johnson Matthey, Inc. (Malvern, PA). Previously described syntheses were used for Rh₂(C₂H₃O₂)₄ (7), Na₄[Rh₂(CO₃)₄] (8), Rh₂(NH₃)₂(C₂H₃O₂)₄ (9), Na[Rh₂(C₂H₃O₂)₄(adenosinemonophosphate)(H₂O)] or Na[Rh₂(C₂H₃O₂)₄(AMP)(H₂O)] (10), and axial bis(amine)complexes of Rh₂(C₂H₃O₂)₄ (9) using ammonia, methyl-butylamine, *n*-propyl-butamine, and *n*-butylamine, where adenosinemonophosphate and alkylamines were purchased from Aldrich Chemical Co. (Milwaukee, WI). Ham's F12 powdered medium came from KC Biological (Lenexa, KS). All chemical solutions were freshly prepared prior to use. All complexes with axial ligands used in these experiments were stored at –75°C. Equilibrium concentrations of species derived from 287 μmol/liter Rh₂(C₂H₃O₂)₄(AMP)(H₂O)[–] in aqueous solution were calculated by the solution of four simultaneous equations appropriately describing this equilibrium as previously reported (11) using the computer program Eureka from Borland International (Scotts Valley, CA).

Electrochemistry. One-electron reduction potentials of 1 mmol/liter Rh₂ complexes in phosphate buffer, pH 7, were determined by cyclic voltammetry using a scan speed of 200 mV/s as previously described (2). Polarography of Na₄[Rh₂(CO₃)₄] was done in phosphate buffer, pH 7, using a Parr Model 174A instrument with silver/silver chloride electrodes.

Pulse radiolysis. Specific rate constants for reactions of Rh₂ complexes in water with the hydrated electron e_{aq}^- were obtained at the Center for Fast Kinetics Research, University of Texas, Austin, Texas, as previously described (2).

Radiation sensitivity. Samples of stationary-phase *Salmonella typhimurium* TM677 at a density of 1×10^8 cells/ml in pH 7 phosphate-buffered saline (PBS), unless otherwise stated, were X-irradiated at a dose rate of 32 Gy/min with N₂ or 100% O₂ gassing as previously described (12), but for one instance where 2% O₂ gassing was used. The fraction of cells surviving graded radiation doses in the presence or absence of 200 μmol/liter Rh₂ complexes was determined with triplicate plating for colony formation. Radiation dose–response curves were constructed, and the radiation sensitivities were determined as the inactivation rate constant k in hGy^{–1}, where hGy is 100 Gy, from the expression $N/N_0 = ne^{-kD}$, where N/N_0 is surviving fraction and D is radiation dose in hGy. The degree of radiation potentiation was determined as the inactivation index (12). Potentiating efficiency for nondose-modifying results was calculated as the threshold dose, i.e., the dose corresponding to the intercept of the N₂ baseline by the extrapolated log-normal response of radiation potentiation.

RESULTS

None of the Rh₂ complexes in this study were toxic under the conditions used. N₂- and 100% O₂-gassed baselines of the radiation dose response were stable and checked frequently and are shown without data points for clarity. Results for all complexes are described at 200 μmol/liter, but for Rh₂(C₂H₃O₂)₄(AMP)(H₂O)[–] at 287 μmol/liter, and this is assumed to be a maximally effective concentration since no difference in response was found for 100 μmol/liter Rh₂ carboxylate complexes (results not shown), and only a slight difference for 100 μmol/liter Rh₂(CO₃)₄(H₂O)₂^{4–} (Fig. 2).

Attempts were made to titrate Na₄[Rh₂(CO₃)₄] in order to quantitate the degree of protonation under conditions of irradiation. Na₄[Rh₂(CO₃)₄] at 2.5 mmol/liter in 0.15 mol/liter N₂-gassed NaClO₄ gave pH 8.7, and titration with HCl initially gave pH changes parallel to those seen in NaClO₄ alone. However, an inflection at pH 6.0 then led to a substantial tailing of the titration curve. Under the same conditions,

titration of 2.5 mmol/liter $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ duplicated the titration curve of NaClO_4 alone. The pH of 200 $\mu\text{mol/liter}$ $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ in N_2 -gassed NaClO_4 solution was 8.0 before and 8.4 after 1 hGy irradiation. The dose-response curves under hypoxic or oxic conditions were not altered when PBS at either pH 8.5 or 7.0 was used.

Rates of reaction with e_{aq}^- in water were found to be essentially diffusion controlled for all carboxylate complexes. These ranged from 1.58×10^{10} liter/mol·s for $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4$ to 2.18×10^{10} liter/mol·s for $\text{Rh}_2(\text{C}_3\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$. This reaction rate was found to remain constant for $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in water and in 0.15 mol/liter NaCl (2). For $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$, the rate of reaction with e_{aq}^- was 4.73×10^8 liter/mol·s, i.e., 40 times less than with the Rh_2 carboxylates or carboxylate amines, but increased to 1.91×10^9 liter/mol·s in 0.15 mol/liter NaCl , i.e., a fourfold increase. Both the constant reaction rate with the neutral Rh_2 complex and the increased reaction rate with the anionic Rh_2 complex were as predicted (13). Results from cyclic voltammetry applied to the axial bis(amine) complexes of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4$ were the same as for the parent diaquo species, i.e., a one-electron reduction potential of -1.1 V (2). The one-electron reduction potential for $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ was found to be -1.6 V using polarography.

Potential of radiation-induced cell killing by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in PBS in the presence and absence of O_2 is shown in Fig. 1. With N_2 gassing the response was biphasic with an insensitive response at low radiation dose followed by a degree of potentiation beyond the threshold radiation dose that is about four times the degree of radiation sensitization seen for O_2 alone, i.e., the ratio of inactivation indices 7/1.62 from Table I. Threshold responses for potentiation were also characteristic of the Rh_2 carboxylates. Threshold doses are listed in Table I and quantitate potentiating efficiency. The degree of potentiation is quantitated in Table I by the inactivation index developed beyond the threshold dose. Unlike the Rh_2 carboxylates, $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ slightly potentiated radiation-induced killing of 100% O_2 -gassed cells, and this without a threshold response. The degree of potentiation by $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in O_2 was 0.04 that of the degree of potentiation in the absence of O_2 . Both the potentiating efficiency and the degree of potentiation decreased as the concentration of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ was reduced from 200 to 10 $\mu\text{mol/liter}$, as shown in Fig. 2, similar to the Rh_2 carboxylates and carboxylate amines (results not shown).

Eliminating phosphate buffer from PBS is known to enhance both the efficiency and the degree of potentiation in the case of Rh_2 carboxylates (2). The results with $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ are shown under similar conditions in Fig. 3, where irradiation in water or in 0.15 mol/liter NaClO_4 rather than in PBS leads to a moderately increased potentiating efficiency and a small increase in the degree of potentiation. Phosphate buffer alone (pH 7) with 9.5 mmol/liter phosphate interfered substantially with the hypoxic radiation potentiation that was seen either in the absence of phosphate buffer or when it was combined with 0.15 mol/liter NaCl as PBS. The use of Ham's F12 media eliminated hypoxic potentiation.

The addition of a variety of amines at a concentration of 2000 to 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in PBS affected radiation potentiation, as shown in Fig. 4. Potentiating efficiency decreased in the order of $n\text{-C}_3\text{H}_7\text{NH}_2 > n\text{-C}_4\text{H}_9\text{NH}_2 > \text{C}_2\text{H}_5\text{NH}_2 = \text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-} > \text{CH}_3\text{NH}_2 > \text{NH}_3$, and the degree of potentiation decreased in the order of $n\text{-C}_4\text{H}_9\text{NH}_2 = n\text{-C}_3\text{H}_7\text{NH}_2 > \text{C}_2\text{H}_5\text{NH}_2 = \text{CH}_3\text{NH}_2$

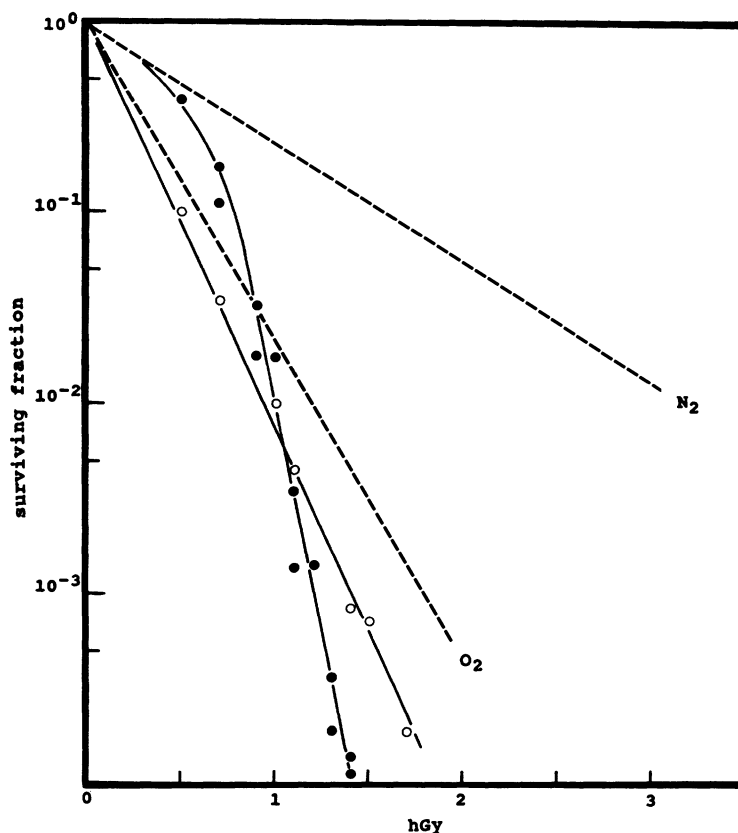


FIG. 1. Hypoxic and oxic radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in PBS. (●) N_2 ; (○) O_2 . Data points of the N_2 and O_2 baselines are omitted.

$> \text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-} > \text{NH}_3$. None of these amines alone at 2000 $\mu\text{mol/liter}$ affected the hypoxic radiation sensitivity (results not shown).

Radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in PBS included a relatively large threshold, but this potentiating efficiency was increased with the use of 200 $\mu\text{mol/liter}$ Rh_2 tetraacetatobis(amine) complexes, as detailed in Fig. 5, where potentiating efficiency decreased in the order $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(\text{NH}_3)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(n\text{-C}_3\text{H}_7\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(\text{CH}_3\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(\text{C}_2\text{H}_5\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$. The degree of potentiation was not greatly affected by these amine ligands, and the decreasing order was generally $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 = \text{Rh}_2(n\text{-C}_3\text{H}_7\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 = \text{Rh}_2(\text{NH}_3)_2(\text{C}_2\text{H}_3\text{O}_2)_4 > \text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2 = \text{Rh}_2(\text{C}_2\text{H}_5\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4 = \text{Rh}_2(\text{CH}_3\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4$. Irradiating cells in the presence of $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4$ and 2% O_2 did not change the response shown in Fig. 5 for that Rh_2 complex, although 2% O_2 alone results in nearly full oxic sensitization (results not shown).

Essentially identical results (not shown) as those above were obtained when 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ was used with the separate addition of 400 $\mu\text{mol/liter}$ of amines. Altering these amine additions to 50 $\mu\text{mol/liter}$ in Fig. 6 or to 2000

TABLE I
Hypoxic Survival Curve Parameters for Rh₂ Complexes and Amines

200 $\mu\text{mol/liter}$ Rh ₂ complex	Threshold dose (hGy)	k_{N_2, Rh_2} ^a (hGy ⁻¹)	Inactivation ^b index
Na ₄ [Rh ₂ (CO ₃) ₄] in PBS	0.67	11	7
Na ₄ [Rh ₂ (CO ₃) ₄] in PBS ^c	—	5	0.3
Na ₄ [Rh ₂ (CO ₃) ₄] in NaClO ₄	0.46	17	10
Na ₄ [Rh ₂ (CO ₃) ₄] in H ₂ O	0.48	25	16
Na ₄ [Rh ₂ (CO ₃) ₄] in PB	ca. 2.7	10	6
In phosphate-buffered saline			
Na ₄ [Rh ₂ (CO ₃) ₄] plus 2000 $\mu\text{mol/liter}$			
NH ₃	1.18	6	3
CH ₃ NH ₂	0.98	16	10
C ₂ H ₅ NH ₂	0.67	17	11
<i>n</i> -C ₃ H ₇ NH ₂	0.45	33	21
<i>n</i> -C ₄ H ₉ NH ₂	0.55	30	19
Rh ₂ (C ₂ H ₃ O ₂) ₄ (H ₂ O) ₂	1.78	16	10
Rh ₂ (NH ₃) ₂ (C ₂ H ₃ O ₂) ₄	0.46	28	18
Rh ₂ (CH ₃ NH ₂) ₂ (C ₂ H ₃ O ₂) ₄	0.93	15	9
Rh ₂ (C ₂ H ₅ NH ₂) ₂ (C ₂ H ₃ O ₂) ₄	1.25	12	7
Rh ₂ (<i>n</i> -C ₃ H ₇ NH ₂) ₂ (C ₂ H ₃ O ₂) ₄	0.54	29	19
Rh ₂ (<i>n</i> -C ₄ H ₉ NH ₂) ₂ (C ₂ H ₃ O ₂) ₄	0.37	33	21
Rh ₂ (C ₂ H ₃ O ₂) ₄ (H ₂ O) ₂ plus 50 $\mu\text{mol/liter}$			
NH ₃	0.77	22	14
CH ₃ NH ₂	1.43	13	8
C ₂ H ₅ NH ₂	1.43	13	8
<i>n</i> -C ₃ H ₇ NH ₂	1.28	30	19
<i>n</i> -C ₄ H ₉ NH ₂	0.61	35	23
Rh ₂ (C ₂ H ₃ O ₂) ₄ (H ₂ O) ₂ plus 2000 $\mu\text{mol/liter}$			
NH ₃	0.54	21	13
CH ₃ NH ₂	0.71	11	6
C ₂ H ₅ NH ₂	0.71	11	6
<i>n</i> -C ₃ H ₇ NH ₂	0.48	38	25
<i>n</i> -C ₄ H ₉ NH ₂	0.48	38	25

^a $k_{N_2} = 1.47 \text{ hGy}^{-1}$ and $k_{O_2} = 3.83 \text{ hGy}^{-1}$ in all suspending media; inactivation index = $(k_{O_2} - k_{N_2})/k_{N_2} = 1.62$.

^b Inactivation index = $(k_{N_2, Rh_2} - k_{N_2})/k_{N_2}$.

^c O₂-gassed suspensions; parameters are inactivation constant = k_{O_2, Rh_2} , and inactivation index = $(k_{O_2, Rh_2} - k_{O_2})/k_{O_2}$.

$\mu\text{mol/liter}$ in Fig. 7 also altered the potentiating efficiencies. In general, decreasing amine concentrations produced decreasing potentiating efficiencies. The degree of potentiation with the above separate amine additions did not change much in order or magnitude compared to the synthesized bis(amine) complexes, but for NH₃ where a somewhat decreased inactivation index was observed.

Rh₂(NH₃)₂(C₃H₅O₂)₄ and Rh₂(*n*-C₃H₇NH₂)₂(C₃H₇NH₂)₂(C₃H₅O₂)₄ were synthesized for comparison with the acetate amine series. The latter complex was unstable in air to the extent that it decomposed to Rh₂(C₃H₅O₂)₄, i.e., its color changed from

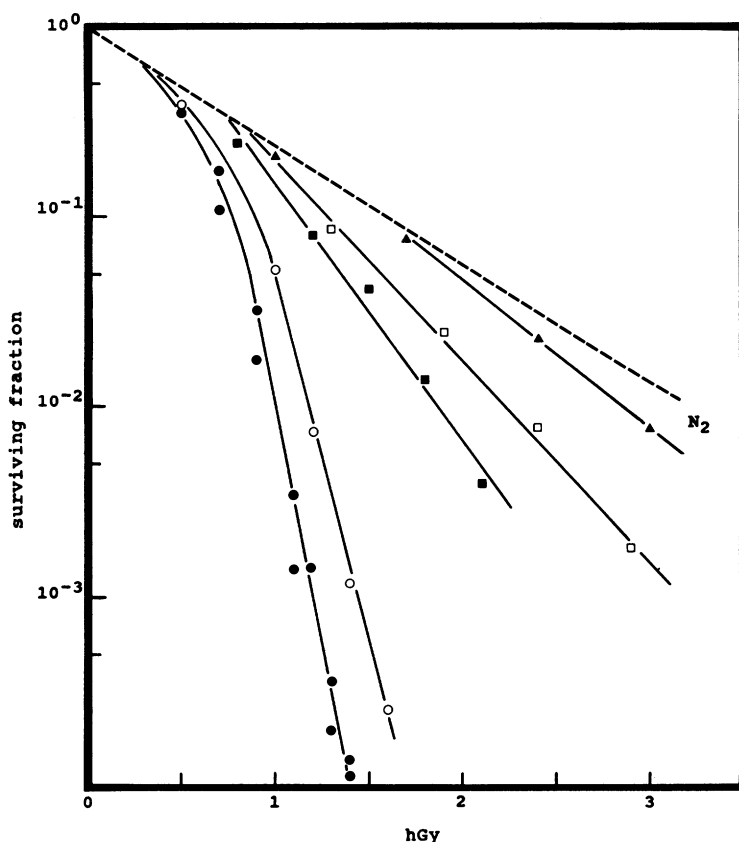


FIG. 2. Hypoxic radiation potentiation by (▲) 10, (□) 25, (■) 50, (○) 100, and (●) 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in PBS.

red to green, during drying after synthesis, and was not studied further. $\text{Rh}_2(\text{NH}_3)_2(\text{C}_3\text{H}_5\text{O}_2)_4$, on the other hand, was stable for months under ambient conditions, and at 200 $\mu\text{mol/liter}$ in PBS potentiated hypoxic cells with a threshold dose = 0.44 hGy, $k = 44.5 \text{ hGy}^{-1}$, and inactivation index = 29.3, i.e., generally comparable to $\text{Rh}_2(\text{NH}_3)_2(\text{C}_2\text{H}_3\text{O}_2)_4$.

The use of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$ at 287 $\mu\text{mol/liter}$ failed to elicit radiation potentiation. Figure 8 shows the absence of potentiation with this complex in hypoxic cells irradiated in water, 0.15 mol/liter NaClO_4 , or in PBS supplemented with 2000 $\mu\text{mol/liter}$ CH_3NH_2 or $n\text{-C}_3\text{H}_7\text{NH}_2$. AMP alone at 287 $\mu\text{mol/liter}$ or combined with 287 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in PBS also failed to potentiate irradiated hypoxic cells (results not shown).

DISCUSSION

Previous investigation of Rh_2 carboxylates in this bacterial system (2) showed decreased radiation potentiation caused by oxygen, decreased potentiating efficiency caused by phosphate, and often similar degrees of potentiation beyond a threshold of

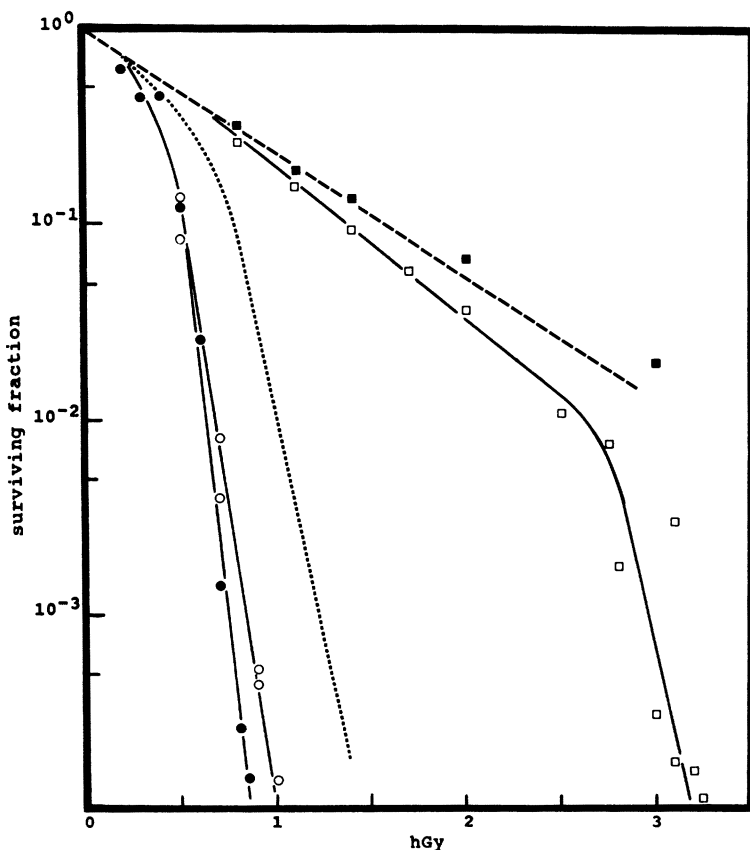


FIG. 3. Effect of suspending media upon hypoxic radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$. (●) water; (○) 0.15 mol/liter NaClO_4 ; (□) 9.5 mmol/liter phosphate buffer; (■) Ham's F12; (.....) PBS; (---) N_2 baseline.

radiation dose for all complexes tested under hypoxic conditions. These findings, coupled with one-electron reduction potentials of -1 V and essentially diffusion controlled rates of reaction with e_{aq}^- for these Rh_2 carboxylates, led to three working hypotheses: (1) one-electron reduction initiated formation of an active radiolytic product common to the different Rh_2 carboxylates; (2) ill-defined phosphate association with the different Rh_2 carboxylates or their one-electron reduced transient in solution interfered with the efficiency of formation of the active radiolytic product; and (3) the active radiolytic product was formed in medium external to the bacterial cell.

The study here examined the above hypotheses by two comparative approaches. First, the carboxylates around the binuclear Rh_2 center were replaced with carbonates to examine the effect of variation of the central cage structure on subsequent radiation potentiation. Second, a series of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4$ complexes was synthesized with amines filling the axial positions, perhaps thereby hindering phosphate association.

Hypoxic potentiating efficiency by $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ at 200 $\mu\text{mol/liter}$ in PBS was 2.7 times greater than by $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$, i.e., 1.78 hGy/0.67 hGy from Table

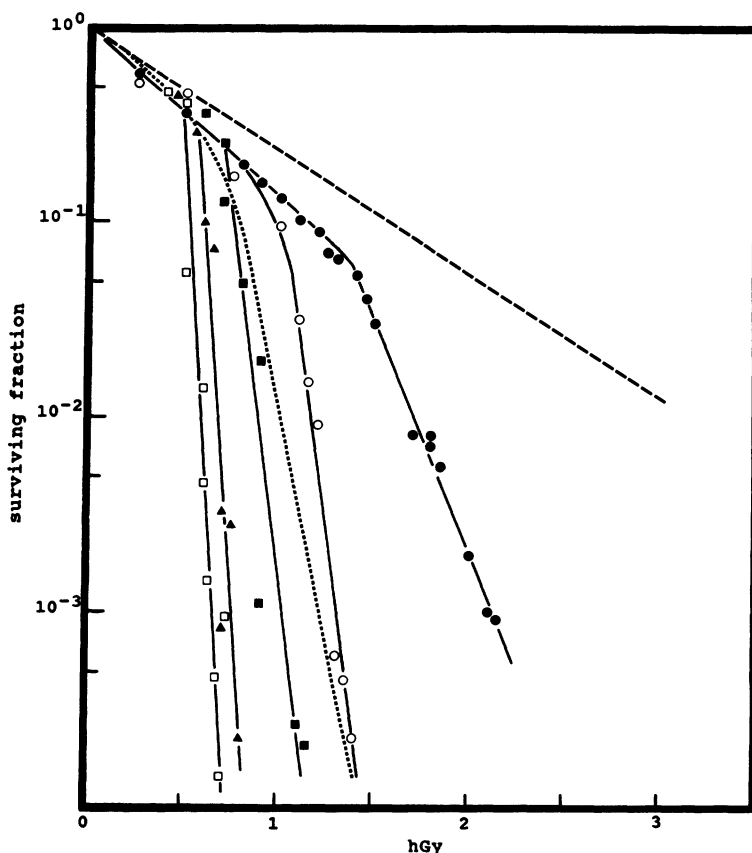


FIG. 4. Hypoxic radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ as affected by addition of 2000 $\mu\text{mol/liter}$ of (●) NH_3 , (○) CH_3NH_2 , (■) $\text{C}_2\text{H}_5\text{NH}_2$, (□) $n\text{-C}_3\text{H}_7\text{NH}_2$, or (▲) $n\text{-C}_4\text{H}_9\text{NH}_2$. Dotted line, $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ alone.

I, and intermediate between $\text{Rh}_2(\text{CHO}_2)_4(\text{H}_2\text{O})_2$ and $\text{Rh}_2(\text{C}_3\text{H}_5\text{O}_2)_4(\text{H}_2\text{O})_2$ that have threshold dose values of 0.16 and 0.98 hGy, respectively (2). The degree of hypoxic potentiation by $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ was substantial in PBS with an inactivation index of 6.5, although this was about two times less than the degree of potentiation by the Rh_2 carboxylates, whose inactivation indices range from 10 to 13 (2). Removal of phosphate from the cell suspension containing $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ led to increased potentiating efficiency and degree of potentiation, with these increases being somewhat less than previously noted for Rh_2 carboxylates (2). Substitution of PBS with phosphate buffer alone, pH 7, decreased the efficiency and the degree of hypoxic potentiation to an extent similar to that with $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ and $\text{Rh}_2(\text{C}_3\text{H}_5\text{O}_2)_4(\text{H}_2\text{O})_2$, while the use of Ham's F12 media eliminated hypoxic potentiation, similar to the case with $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ (2). To this point, therefore, comparisons of radiation potentiation caused by $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ and the Rh_2 carboxylates are similar.

$\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ reacts 40 times less rapidly with e_{aq}^- in water and 10 times less rapidly with e_{aq}^- in 0.15 mol/liter NaCl than Rh_2 carboxylates (2). The potentiating

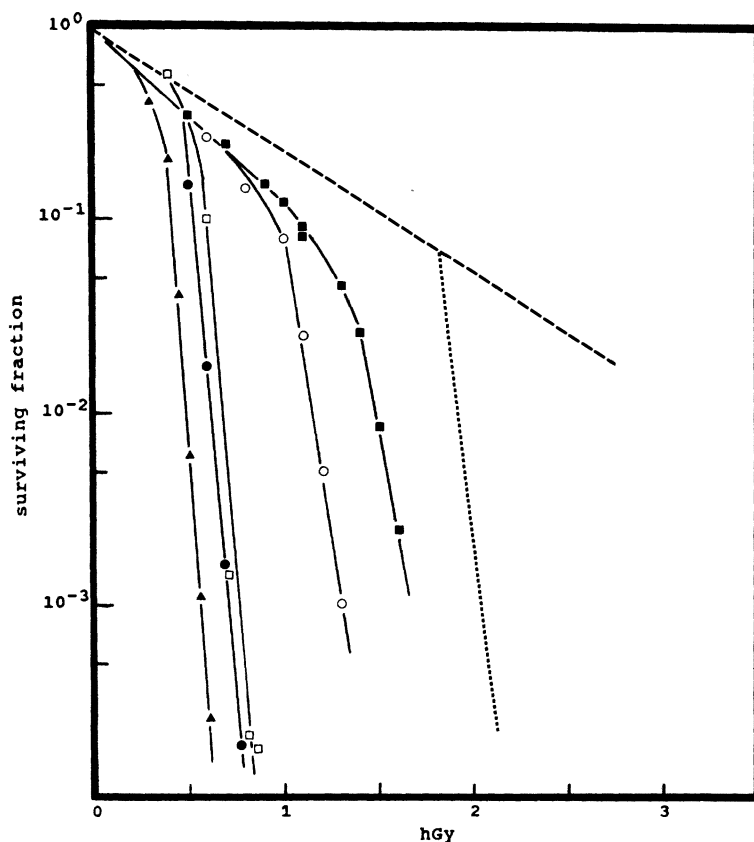


FIG. 5. Hypoxic radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2\text{L}_2(\text{C}_2\text{H}_3\text{O}_2)_4$, where L is (●) NH_3 , (○) CH_3NH_2 , (■) $\text{C}_2\text{H}_5\text{NH}_2$, (□) $n\text{-C}_4\text{H}_7\text{NH}_2$, or (▲) $n\text{-C}_4\text{H}_9\text{NH}_2$. Dotted line, $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4$ alone.

efficiency of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ relative to Rh_2 carboxylates might therefore decrease in water or in 0.15 mol/liter NaClO_4 since one-electron reduction of Rh_2 complex is hypothesized to form the active product. The threshold doses for $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in water or in NaClO_4 (Table I) were greater than for the Rh_2 carboxylates involving formate, acetate, and propionate (2). Although these differences in potentiating efficiencies did not approximate the 40-fold or 10-fold differences in the e_{aq}^- reaction rate, neither does the reaction rate alone predict the final yield of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ reacted with e_{aq}^- . Competition for e_{aq}^- is limited in these cell suspensions, presumably then allowing $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ as the primary reactant with e_{aq}^- .

The negative charge of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ is presumed to be sufficient to exclude it from the bacterial protoplast. This negative charge reflects the degree of protonation, that in turn reflects the pH of the solution. The titration curve of $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ was unexpectedly complex below pH 6, so that the proton equilibria could not be determined formally. However, 200 $\mu\text{mol/liter}$ $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ in unbuffered N_2 -gassed 0.15 mol/liter NaClO_4 gave a pH of 8.0, indicating only a small degree of monoprotation. These results are taken as support of the hypothesis that

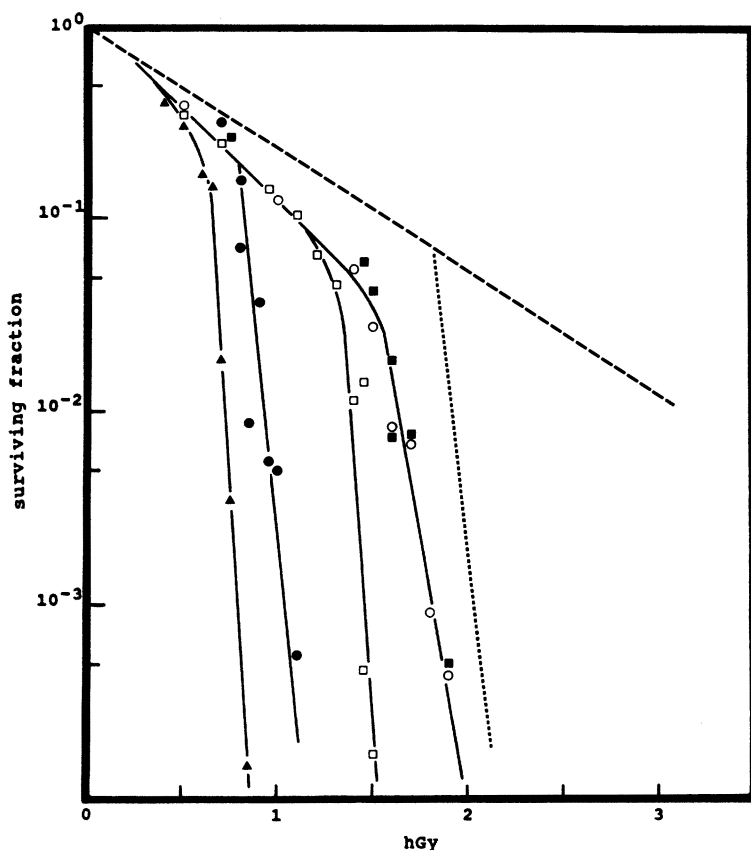


FIG. 6. Hypoxic radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ as affected by addition of 50 $\mu\text{mol/liter}$ of (●) NH_3 , (○) CH_3NH_2 , (■) $\text{C}_2\text{H}_5\text{NH}_2$, (□) $n\text{-C}_3\text{H}_7\text{NH}_2$, or (▲) $n\text{-C}_4\text{H}_9\text{NH}_2$. Dotted line, $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ alone; dashed line, N_2 baseline.

the initial radiochemical event leading to radiation potentiation by these Rh_2 complexes occurs in solution external to the bacterial cell membrane (2).

Radiation potentiation by $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ deviates from that of the carboxylates in two ways that may reflect some variation in the mechanism of action. The first is the occurrence of a small degree of radiation potentiation with 100% O_2 gas-sing, and the second is a difference in the profile of potentiation in the presence of various amines compared to $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$.

In the first case, O_2 at 1.3 mmol/liter, and with a reaction rate for e_{aq}^- of 2×10^{10} liter/mol $\cdot$ s (14), will scavenge essentially all e_{aq}^- from 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$. Subsequent formation of active product is considered unlikely for two reasons. First, electron transfer to Rh_2 from O_2^- is unlikely due to the low one-electron reduction potential of this Rh_2 complex. Second, adduct formation of Rh(II) with O_2^- might occur (15), but resultant build-up of the active product is not indicated since no threshold dose was seen in 100% O_2 (Fig. 1). An enhanced sensitization by O_2 is suggested.

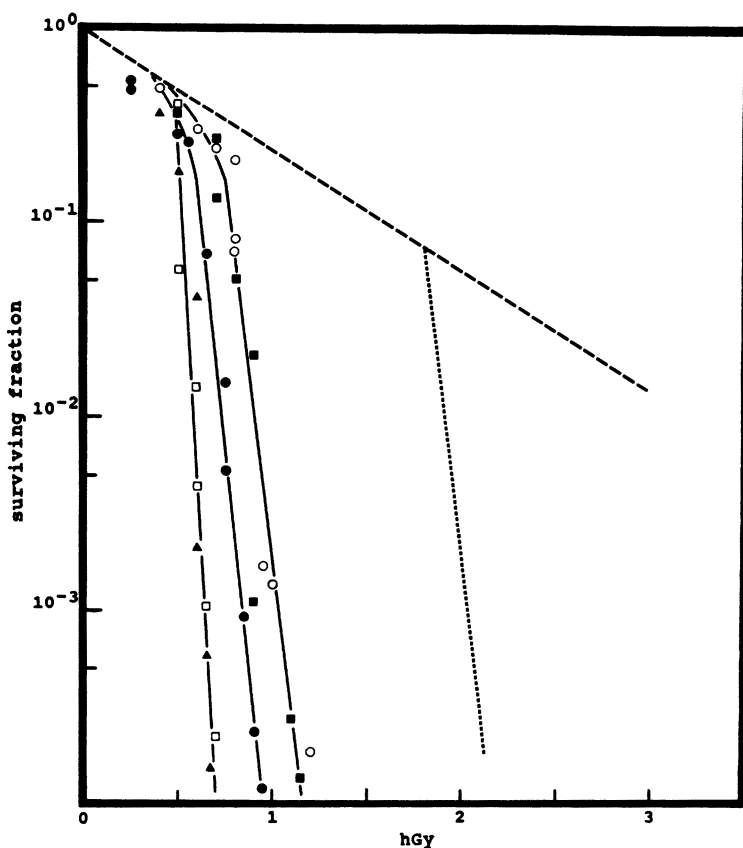


FIG. 7. Radiation potentiation by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ as affected by addition of 2000 $\mu\text{mol/liter}$ of (●) NH_3 , (○) CH_3NH_2 , (■) $\text{C}_2\text{H}_5\text{NH}_2$, (□) $n\text{-C}_3\text{H}_7\text{NH}_2$, or (▲) $n\text{-C}_4\text{H}_9\text{NH}_2$. Dotted line, $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ alone; dashed line, N_2 baseline.

The existence of oxidic potentiation by $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ contrasts with the Rh_2 carboxylates where 100% O_2 gassing results in only the O_2 baseline response. For the latter observation, O_2 is proposed to scavenge e_{aq}^- , thus preventing formation of the active radiolytic product. This proposal is supported by the finding that at 2% O_2 the radiation potentiation caused by 200 $\mu\text{mol/liter}$ $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4$ is identical to the hypoxic potentiation shown in Fig. 5. Alone, 2% O_2 caused 70% of the full oxidic response. Additionally, oxygen could complex one-electron reduced Rh_2 transients (15) or reoxidize this transient by an electron transfer process (1), thereby preventing the development of radiolytic product otherwise obtained.

In the second case, the presence of 2000 $\mu\text{mol/liter}$ NH_3 or CH_3NH_2 decreased the potentiating efficiency of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in PBS, and NH_3 reduced the degree of potentiation by a factor of 2, i.e., the ratio of inactivation indices 6/3 from Table I. This behavior contrasts with the increased potentiating efficiency caused by 2000 $\mu\text{mol/liter}$ NH_3 or CH_3NH_2 with $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$; NH_3 slightly increased the degree of potentiation, as shown in Table I. These amines have an incomplete affinity, as discussed below, for the axial positions of both the Rh_2 carbonate and carboxylate

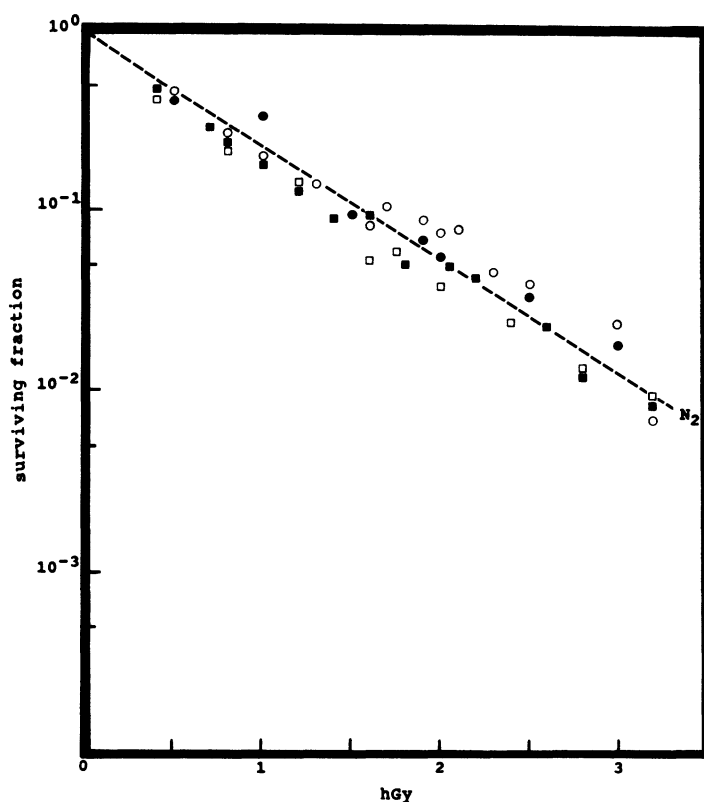


FIG. 8. Absence of hypoxic radiation potentiation by $287 \mu\text{mol/liter Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$ in (●) $0.15 \text{ mol/liter NaClO}_4$, (○) water, (■) $2 \text{ mmol/liter CH}_3\text{NH}_2$ and PBS, or (□) $2 \text{ mmol/liter } n\text{-C}_3\text{H}_7\text{NH}_2$ and PBS.

complexes in dilute aqueous solution. Therefore, differences in potentiation with changing amine concentrations are proposed to relate to axial ligand chemistry. However, no specific hypothesis is offered for these differences at this time.

Rh_2 carboxylates are blue-green, whereas axial coordination to nitrogen produces pink to red (11). This color pertains to both the complexes and their aqueous solutions. The bis(amine) complexes of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4$ used in Fig. 5 were pink to red, but at $200 \mu\text{mol/liter}$ their solutions were blue-green. Increasing amine concentrations to $2000 \mu\text{mol/liter}$ introduced a tinge of red. Similar observations of color were made for solutions of $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$.

Obviously, dissociation of the axial amines occurs to give substantial $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in dilute aqueous solution. Such dissociation likely explains the one-electron redox potentials of all bis(amine) complexes of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4$ being equal to the diaquo parent species. Nonetheless, the effect of the amines was to increase the efficiency of potentiation by $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in PBS, approaching the potentiating efficiencies seen for $200 \mu\text{mol/liter Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in the absence of phosphate.

Protonated alkyl amines, i.e., pH 7, do not react well with the primary yields of water radiolysis (16). It is suggested that the dynamic equilibrium of amines with

$\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ interferes with phosphate associations that hinder the formation or function of the active radiolytic product.

The amine enhancement of potentiating efficiency was also tested in the tetrapropionate system by comparing 200 $\mu\text{mol/liter}$ of $\text{Rh}_2(\text{C}_3\text{H}_5\text{O}_2)_4(\text{H}_2\text{O})_2$ and $\text{Rh}_2(\text{NH}_3)_2(\text{C}_3\text{H}_5\text{O}_2)_4$ in PBS. For this amine complex, threshold dose decreased (0.98 to 0.43 hGy, respectively) and inactivation rate constant increased (17 to 38 hGy^{-1} , respectively). Therefore, comparative findings agree with the Rh_2 tetraacetate system.

The lack of potentiation by $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$ in Fig. 8 is significant compared to the substantial potentiation observed for the Rh_2 carboxylate amines. Like these, $\text{Na}[\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})]$ is red but becomes blue-green at 287 $\mu\text{mol/liter}$ in aqueous solution, indicating dissociation to aquated species. Formation constants for $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})_2^{2-}$ and for $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$ in water have been reported as 202 and 1893, respectively (17). Equilibrium concentrations of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})_2^{2-}$, $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$, $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$, and AMP were therefore calculated as 3, 79, 205, and 202 $\mu\text{mol/liter}$, respectively, for the initial 287 $\mu\text{mol/liter}$ of $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})^-$ (Fig. 8). Knowing these equilibrium concentrations, it can be hypothesized that the absence of potentiation by $\text{Na}[\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})]$ is due to the inactivation by AMP of the pathway of radiolytic product formation from $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$, as follows. Competition for e_{aq}^- in the solution of $\text{Na}[\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{AMP})(\text{H}_2\text{O})]$ only diverts about 39% of the e_{aq}^- yield (17) from reaction with 205 $\mu\text{mol/liter}$ $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$. This condition is more than sufficient for full potentiation by $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ (2). Yet no potentiation was observed in PBS, NaClO_4 , or water, or in the presence of two different amines (Fig. 8). Since the interference with the reductive event forming the one-electron reduced Rh transient is not sufficient, it remains that the subsequent disposition of this transient may be hindered by the presence of AMP, e.g., by oxidative restitution of the one-electron reduced Rh_2 transient (1).

In summary, carbonates comprising the ligand cage of $\text{Rh}(\text{II})_2$ caused a 40-fold decreased reactivity with e_{aq}^- in aqueous solution compared to carboxylates, that in turn correlated with a slightly decreased degree of potentiation. One-electron reduction of $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ external to the cell leading then to an active product in a pathway that is sensitive to phosphate interference is hypothesized for hypoxic radiation potentiation by this complex as well as by the Rh_2 carboxylates (2). The small increase in radiation sensitivity seen for $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ in the presence of 100% O_2 is poorly understood, but may reflect increased sensitization by O_2 rather than potentiation by the Rh_2 complex. The noninhibition at 2% O_2 compared to the full inhibition at 100% O_2 of the hypoxic potentiation seen for 200 $\mu\text{mol/liter}$ $\text{Rh}_2(n\text{-C}_4\text{H}_9\text{NH}_2)_2(\text{C}_2\text{H}_3\text{O}_2)_4$ is suggested to be due to the competition by O_2 for e_{aq}^- , further supporting the one-electron reduction pathway hypothesized to lead to active product and Rh_2 -based radiation potentiation. Inclusion of amines with $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ in PBS is hypothesized to decrease interference by phosphate on hypoxic potentiation, perhaps via dynamic equilibrium of the amines at the axial positions of the Rh_2 carboxylate. The same amines included with $\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2^{4-}$ affected hypoxic potentiation to give a profile different than with $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$. Some mechanism-related differences between these latter two complexes are therefore allowed, despite otherwise comparable results. AMP as an axial ligand to $\text{Rh}_2(\text{C}_2\text{H}_3\text{O}_2)_4(\text{H}_2\text{O})_2$ prevented potentiation, despite substantial disso-

ciation in solution, suggesting inhibition by AMP of active product derived from the one-electron reduced Rh₂ pathway.

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